

Experimental Demonstration of Autonomous Class Aware Optimization via Digital Twin Aided Optical Network Agent

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Abstract We demonstrate a Digital Twin-aided multi-agent LLM for autonomous class-aware optical network management on a network testbed. The system detects soft failures and autonomously adapts modulation format per service class requirements, ensuring marginless channel quality with minimal human intervention. ©2026 The Author(s)

Introduction

Modern operational demands of optical networks are driving the development of novel, domain-specific LLM models and Agentic frameworks for network intelligence [1]. These models enable closed-loop control, intent-driven operation, and real-time decision-making [2], shifting network management from manual/rule-based approaches to autonomous, AI-native control. These functionalities are fundamental in the context of marginless operations [3], where a network is pushed more towards its limits, requiring proper performance observation and reactions in case of problems. Recent demonstrations highlight the feasibility of LLM-powered agents for autonomous driving optical networks to manage wavelength add/drop operations, failure detection, and perform power optimization across the network lifecycle [4].

Further work shows that multi agent LLMs can enhance digital-twin-based workflow automation by integrating intent parsing, telemetry analysis, auto-simulation, and closed-loop optimisation, enabling QoT-aware provisioning and self-improving reasoning in operational environments [5]. These advances illustrate the growing role of multi-agent LLM architectures in decomposing complex optical network tasks into specialized roles - planning, telemetry interpretation, configuration execution, and validation - improving reliability and reducing human intervention.

To the best of the authors' knowledge this work is the first to experimentally demonstrate a statistical Digital Twin (DT)-assisted SDN controller employing a multi-agent framework for seamless physical and network layer integration in autonomous optical service optimization.

We benchmark and demonstrate an optical domain fine-tuned LLM for reasoning and decision-making. LLaMA3.1-8B fine-tuned performs best across all evaluated models. In a multi-agent scenario, supported by a DT, the fine-tuned LLM reliably executes QoT-driven decisions for marginless networking. We experimentally

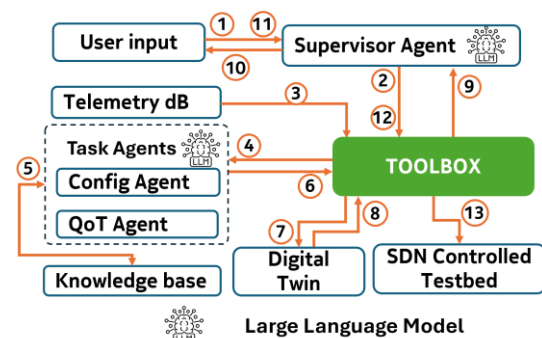


Fig. 1: Proposed multi-agent LLM architecture

validate that the system autonomously detects soft failures before they escalate and leverages DT-assisted QoT predictions to maintain fully automated operation of marginless connections over a 630 km path under emulated ageing.

Multi-Agent LLM Architecture

The control architecture shown in Fig. 1 instantiates a plan–verify–execute workflow. Three LLM-based agents are present in the architecture: Supervisor, Config, and QoT agents. The workflow is initiated at the human–machine interface, where natural language input (1) is received by the Supervisor Agent — a fine-tuned LLaMA3.1-8B responsible for intent parsing and task delegation. The Supervisor forwards the interpreted task (2) to the Toolbox, a centralized orchestration bus that simultaneously ingests real-time telemetry (3) from a NETCONF/YANG-pollled Telemetry Database, ensuring all downstream actions are grounded in current network state. The Toolbox dispatches tasks (4) to two specialized agents: the Config Agent translates high-level intent into vendor-neutral NETCONF remote procedure calls (RPCs) for amplifier gain, variable optical attenuator (VOA), and wavelength selective switch (WSS) settings; the QoT Agent performs quality-of-transmission estimation and parameter optimization based on detected impairments.

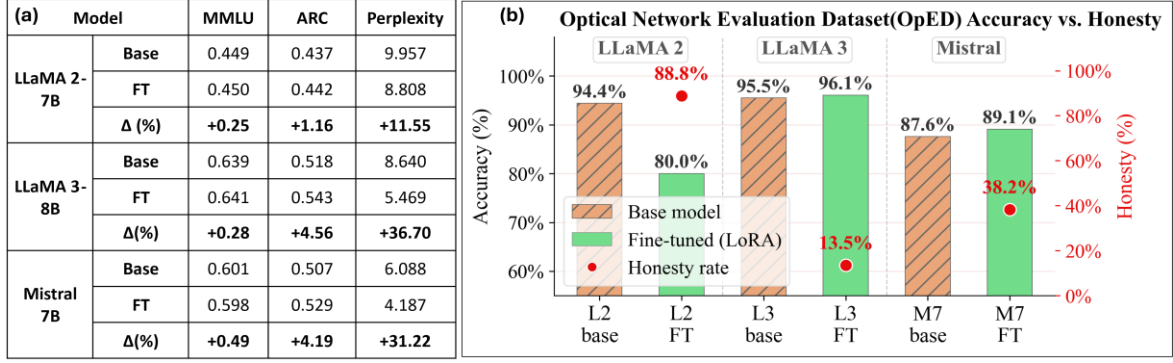


Fig. 2: (a) Standard benchmarking evaluation - base and fine-tuned (FT) LLMs; (b) Evaluation of base and fine-tuned LLMs

Both are fine-tuned LLaMA3.1-8B models that query (5) a shared Knowledge Base - containing standards, vendor specifications, and physical-layer constraints - via Retrieval-Augmented Generation (RAG), an inference-time mechanism that conditions model outputs on retrieved domain context to suppress hallucination [6]. Agent responses are returned (6) to the Toolbox.

Before any configuration is applied, the Toolbox forwards candidate actions (7) to the Digital Twin, which validates them against physical-layer constraints - gain limits, margin requirements, and forward error correction (FEC) thresholds - and returns a statistical validation (8). The outcome is reported back (9) to the Supervisor Agent, which communicates the decision (10) to the user; if further iteration is required, the loop re-enters (11, 12) until a valid action is confirmed. Only verified configurations are dispatched (13) to the SDN controller ensuring physically inconsistent or overconfident LLM outputs are intercepted before reaching live hardware.

Three models — LLaMA2-7B [7], LLaMA3.1-8B [8], and Mistral-7B [9] — were fine-tuned via QLoRA (Quantized Low-Rank Adaptation) [10] on testbed telemetry and ITU-T standards documentation. They were first evaluated on standard LLM benchmarks: MMLU (Massive Multitask Language Understanding), a set of multiple-choice questions [11], ARC (AI2 Reasoning Challenge), designed to evaluate reasoning capabilities [12], and perplexity, a measure of how accurately a model predicts the next token in a sequence [13]. Second, the models were assessed on OpED — the Optical Network Evaluation Dataset — a domain-specific benchmark developed in this work, covering optical hardware, DSP, and SDN/NFV automation. OpED additionally applies an honesty penalty for hallucinated responses. LLaMA3.1-8B fine-tuned achieved the strongest results across both evaluation axes: the largest perplexity reduction at 36.70% on standard benchmarks (Fig. 2a), and the highest OpED accuracy at 96.1% (Fig. 2b). Its low honesty rate of 13.5% reflects a small number of abstained

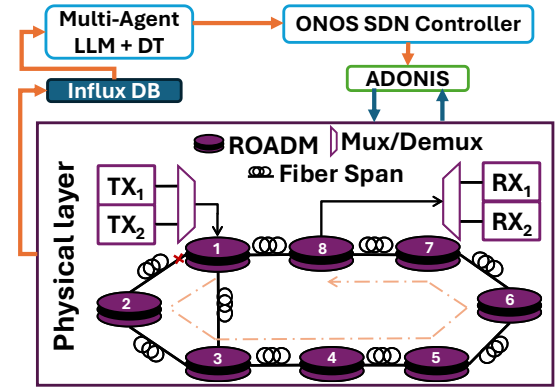


Fig. 3: Experimental testbed topology

questions, the majority of which were answered correctly unlike base models, which attempted all questions regardless of confidence and yielded no honesty signal. Given its overall superiority, the fine-tuned LLaMA3.1-8B was selected for all agent roles.

Experimental demonstration and results

The physical layer testbed in Fig. 3, consists of 8 commercial-grade reconfigurable add/drop multiplexer (ROADM) nodes, connected in a mesh topology. We transmit two 300G channels from different commercial transceivers in a 175 GHz shared slot, operating under two different service class agreements (SLA): *gold class* and *bronze class*. Gold SLA is to maintain constant net rate of 300G with a high margin, ensuring robust service delivery. Conversely, Bronze SLA allows the channel to operate at zero-margin, with the caveat that the net rate can be ceded under strenuous service conditions to satisfy the Gold SLA.

The testbed has continuous monitoring of the pre-FEC BER of both channels along with the OSNR at each node, total spectrum and channel powers at the input and output of each node, amplifier gains, VOA and WSS attenuations from the ROADMs. The telemetry data is stored in the InfluxDB and is visualized on the Grafana interface. We use an ONOS-based SDN controller for the data plane. The multi-Agent framework employs Python functions to handle analytics while the

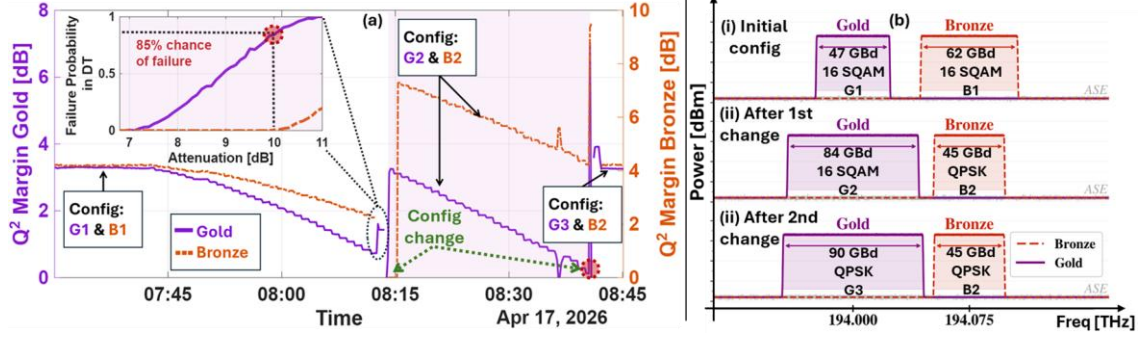


Fig. 4: (a) Timeseries of mode transitions for gold and bronze service after verification of performance within the DT, and failure probability predicted by the twin; (b) Optical spectrum, before and after service-based mode adaptation.

LLM provides higher level reasoning.

For soft-failure tracking, the QoT Agent compares the pre-FEC BER to a threshold $Th_{SF} = \langle BER \rangle + 4\sigma$ (from a 9-sample window and standard deviation σ obtained from live measurements) triggering adaptation when BER reaches about 80% of the FEC margin, defined in Q-factor calculated from the measured pre-FEC BER [14] as: $Q^2_{margin} = Q^2_{measured} - Q^2_{FEC}$ (dB). A failure detection algorithm selects either a lower-rate or lower-modulation mode and evaluates feasibility within a DT. The DT exploits past measurements to emulate statistical network performance under the proposed failure trend, following the method in [10]. It returns the expected Q-factor along with failure probability corresponding to $Q^2_{margin} < 0$.

The assessment and cross-channel context are passed to the LLM, which issues an APPROVE/DEFER/REJECT decision. If approved, the Configuration Agent deploys the new mode via the ONOS SDN Controller.

To evaluate robustness, fiber aging is emulated by increasing VOA attenuation at ROADM 1 from 0 dB to 12 dB in 0.2 dB steps, each held for 7 minutes. The agentic workflow is initialized via the user prompt: *"in case of soft failure, run modulation-format adaptation"*. At 0 dB, LLM agents establish a baseline BER and compute the Th_{SF} . As attenuation increases, the Q-margin narrows for both Gold and Bronze services.

The mode-adaptation algorithm is triggered when, at 6.8 dB attenuation, the BER approaches ~80% of the Q^2_{margin} . To preserve the Gold SLA of 300G net rate, the Configuration Agent proposes downgrading the Bronze service from B1 (300G) to B2 (100G), freeing sufficient spectrum. Before deployment, the DT validates this configuration by predicting the Q^2_{margin} and failure probability for both services, confirming SLA compliance. Fig. 4(a) captures the Gold service mode transition from G1 to G2, and Fig. 4(b) shows the resulting spectral reallocation.

As attenuation further increases to 10 dB, the DT-calculated failure probability for the Gold

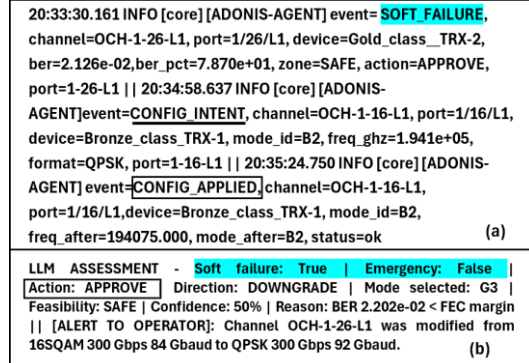


Fig. 5: (a) ONOS SDN logs for configs; (b) LLM assessment with decision rationale and operator alert

service reaches 85% (Fig. 4(a), inset), necessitating a second transition from G2 to G3, while B2 is retained unchanged. Throughout both transitions, the validated configurations are deployed via ONOS, demonstrating the closed-loop adaptability of the proposed framework under progressive link degradation.

Fig. 5(a) presents the ONOS event log capturing the full agent event sequence — SOFT_FAILURE detection, CONFIG_INTENT generation, and CONFIG_APPLIED confirmation — during live testbed reconfiguration. Fig. 5(b) reports the corresponding LLM assessment during the G2 to G3 transition, detailing the decision rationale, confidence level, and operator alert issued upon mode adaptation.

Conclusion

This work evaluates fine-tuned LLMs for autonomous optical network management, with LLaMA3.1-8B-FT emerging as the top-performing model across all standard benchmarks. Integrated within a statistical Digital Twin aided multi-agent framework, the proposed approach is experimentally validated on a network testbed with two separate classed services, demonstrating closed-loop, zero-touch mode adaptation with sustained service level agreement compliance under emulated fiber ageing.

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